

Edge Detector

This project implements a simple edge detection application in Python. The application applies Gaussian smoothing followed by Sobel operators to compute horizontal and vertical image gradients and the gradient magnitude. A Tkinter GUI allows the user to load an image, adjust the Gaussian kernel size and sigma, and visualize the results.

GUI Examples

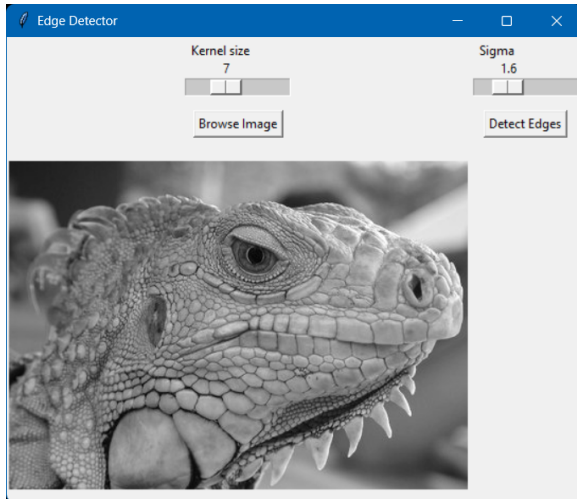


Figure 1. Iguana GUI

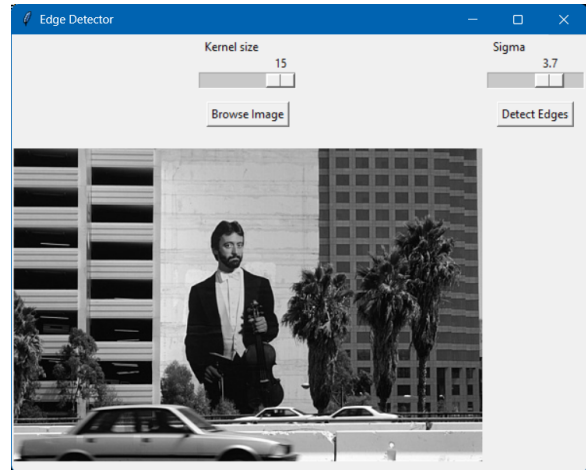


Figure 2. Building GUI

Edge Detection Results

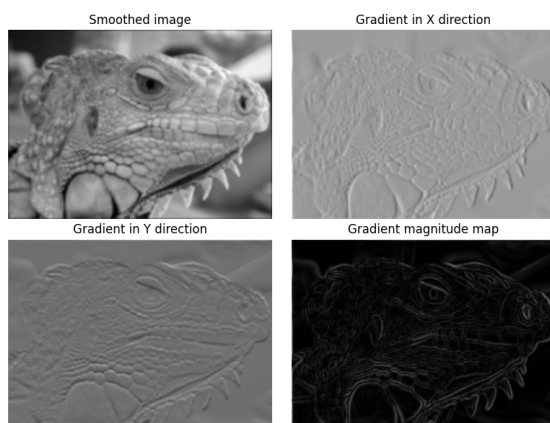


Figure 3. Iguana Results

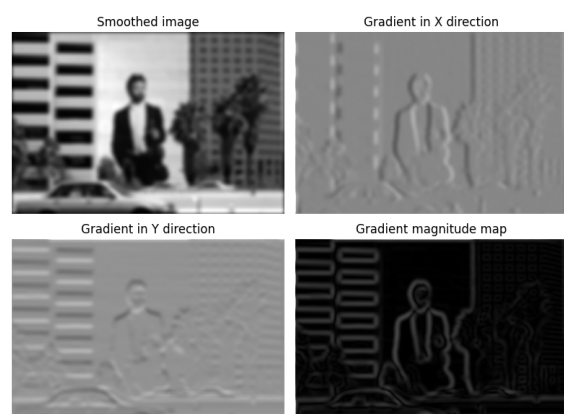


Figure 4. Building Results